

SESSION – 13

Java Priority Queue

```
import java.util.*;

class Student {
    private int id;
    private String name;
    private double cgpa;

    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    public int getID() {
        return id;
    }

    public String getName() {
        return name;
    }

    public double getCGPA() {
        return cgpa;
    }
}

class Priorities {

    public List<Student> getStudents(List<String> events) {

        PriorityQueue<Student> pq = new PriorityQueue<>(
            (a, b) -> {

                int result = Double.compare(b.getCGPA(),
a.getCGPA());

                if (result != 0) {
                    return result;
                }

                result = a.getName().compareTo(b.getName());
            }
        );
    }
}
```

```

        if (result != 0) {
            return result;
        }

        return Integer.compare(a.getID(), b.getID());
    }
};

for (String event : events) {

    String[] parts = event.split(" ");

    if (parts[0].equals("ENTER")) {

        String name = parts[1];
        double cgpa = Double.parseDouble(parts[2]);
        int id = Integer.parseInt(parts[3]);

        pq.add(new Student(id, name, cgpa));

    } else if (parts[0].equals("SERVED")) {

        if (!pq.isEmpty()) {
            pq.poll();
        }
    }
}

List<Student> result = new ArrayList<>();

while (!pq.isEmpty()) {
    result.add(pq.poll());
}

return result;
}
}

public class Solution {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        sc.nextLine();
    }
}

```

```

List<String> events = new ArrayList<>();

for (int i = 0; i < n; i++) {
    events.add(sc.nextLine());
}

Priorities priorities = new Priorities();

List<Student> students = priorities.getStudents(events);

if (students.isEmpty()) {
    System.out.println("EMPTY");
} else {
    for (Student student : students) {
        System.out.println(student.getName());
    }
}

sc.close();
}
}

```

Congratulations

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Test case 0

Test case 1

Test case 2

Test case 3

Test case 4

Test case 5

Test case 6

Compiler Message

Success

Input (stdin)

1 12
2 ENTER John 3.75 50
3 ENTER Mark 3.8 24
4 ENTER Shafaet 3.7 35
5 SERVED
6 SERVED
7 ENTER Samiha 3.85 36
8 SERVED

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